

Enabling Commercial Lunar Transportation to Support a Sustained Lunar Base

**National Aeronautics and Space Administration (NASA)
George C. Marshall Space Flight Center
Huntsville, AL 35812**

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Responses Due: April 23, 2026

1. SUMMARY

NASA is seeking information from established commercial providers and new industry entrants to help shape a future acquisition strategy for Artemis Commercial Lunar Transportation Solutions that deliver the human cadence required to establish and sustain a permanent lunar base.

This RFI is aligned with the Executive Order “Ensuring American Space Superiority” (December 2025), which directs the United States to:

- Establish a permanent, continuously crewed U.S. lunar base
- Leverage rapid, resilient, commercially provided space infrastructure to maintain U.S. leadership in the cislunar domain

Consistent with the EO’s call for accelerated human mission cadence, high tempo space operations, commercial partnership, and reliable interoperability with current and future elements of the Artemis architecture, NASA is seeking transportation concepts capable of supporting the sustained buildup, logistics resupply, and crew rotation tempo required for lunar base operations.

Proposed human transportation systems must be capable of rendezvous, proximity operations, and docking (RPOD) with existing Human Landing System (HLS) providers. Proposed architectures must also demonstrate a clear path to supporting future Artemis architecture requirements.

No specific Earth or lunar orbit rendezvous location is prescribed. NASA encourages providers to propose trajectories, staging orbits, and operational profiles that best support both current HLS vehicles and future Artemis needs.

NASA welcomes input from:

- Incumbent providers, particularly solutions that evolve existing systems to increase cadence, reduce cost, and accelerate readiness, and transition toward commercially sustainable human space transport capabilities; and
- New entrants, including innovative architectures that expand the commercial space industrial base. All new entrants must describe a credible path forward toward human-rating certification, including how their architectures, development plans, and programmatic approaches address the requirements of a human-class transportation system.

Optional additional capabilities that extend long-term architectural value may be included, provided they remain clearly secondary to the primary objective of enabling human lunar transportation services.

2. BACKGROUND

NASA's long-term Moon-to-Mars strategy includes establishing a permanent, continuously crewed lunar base that supports science, surface operations, technology maturation, and industrial activity. To accelerate this vision, NASA has standardized Artemis vehicle configurations through Artemis V, allowing increased human-mission tempo and reduced developmental risk. NASA now seeks industry perspectives on how incumbent systems can evolve and how new architectures can be introduced to support future human transportation needs. To achieve long-term surface exploration goals and to enable future commercial activity on the lunar surface, NASA seeks Industry's input on approaches to evolve the current architecture to solutions that provide commercial transportation for Artemis missions and have synergy with commercial objectives.

Executive Order Policy Driver

The Executive Order "*Ensuring American Space Superiority*" establishes the lunar region as a strategic national interest and directs the United States to:

- Establish a sustained human presence on the Moon
- Deploy supporting lunar space infrastructure on an accelerated schedule
- Ensure U.S. capabilities are resilient, competitive, and commercially leveraged
- Maintain leadership in the cislunar operating volume, ensuring U.S. freedom of action

This policy mandates a transportation architecture capable of high-frequency mission operations, reliability, redundancy, and commercial scalability.

To ensure compatibility with ongoing Artemis missions and future lunar surface operations, all human rated transportation systems must:

- Rendezvous and dock with existing HLS providers, enabling crew transfer to and from the lunar surface
- Maintain interoperability with evolving Artemis elements, including HLS landers and docking standards.

For current human rated transportation systems proposed to be used, articulate approach to adapt and evolve mission architectures without redesigning or re-certifying current human-rated transportation systems.

For proposed new transportation systems, specifically articulate path to achieving human-rating certification, including plans to meet safety, reliability, verification, and human-systems integration requirements during development and operations.

Respondents may describe optional capabilities that extend architectural value or future mission utility, provided these are clearly secondary and do not shift the primary focus away from human transportation services.

3. INFORMATION SOUGHT

Assumptions of NASA's Desired Future State:

- Two commercial providers for end-to-end lunar transportation
 - For this RFI, 'end-to-end' refers to transportation from Earth launch through arrival to an orbital transfer location (e.g., NRHO or alternative staging orbit), and return from the orbital transfer location to Earth.
 - Surface delivery is not within scope except insofar as it informs docking or transfer operations with HLS providers.
- Crewed missions delivered to surface every 6 months (1 mission per year for each provider, initially)
- Respondents should provide sensitivity of launch cadence and surface duration, and their effects on unit costs.

Respondents may address any or all of the following topics:

A. Technical Topics

General:

1. Describe your end-to-end lunar transportation architecture for human missions, including optional support for additional mission classes where such capabilities do not detract from the primary human-transport objective.
2. Identify achievable launch and mission cadence, including paths to scale to EO driven requirements.

3. Proposed Earth and lunar operational orbits, trajectories, and staging concepts—no specific orbit is required.
4. Identify the crew transfer location(s) your architecture assumes within the Earth–Moon system and discuss operational and cadence implications of your chosen location(s).
5. Describe autonomous, loiter, quiescent, or dormant operations for multi mission campaigns.
6. Discuss the feasibility and impacts of providing manual crew control capability during selected mission phases, including implications for system design, safety, operations, and certification.
7. Describe the capability, if any, of your transportation architecture to accommodate secondary scientific payloads or technology demonstration payloads on crewed or uncrewed missions.

Incumbents:

1. Describe evolution strategies from the current Artemis architecture to a commercially sustainable human-transportation architecture, including growth paths for increased capabilities and maturation of key technologies affecting schedule or integration risks.

New Entrants:

1. Identify maturity levels of system elements and key enabling technologies, including associated low-TRL items and your mitigation strategies.
2. Explain your safety, abort strategy, and hazard control approaches.
3. Describe the certification basis you anticipate using (e.g., NASA standards, commercial human-spaceflight frameworks, or a hybrid approach) and identify key technical or programmatic risks associated with meeting those requirements.

Interoperability with HLS and Artemis (Required for Human Missions)

Describe how your system will:

1. Perform rendezvous, proximity operations, and docking with existing HLS providers.
2. Accommodate current HLS interface standards.
3. Maintain interoperability with future Artemis architecture needs.

4. Support evolving lander classes, docking geometries, and mission designs.
 5. Offer optional enhancements that improve interoperability, reduce risk, or increase cadence
 6. Describe any software or hardware test, emulation, or integration capabilities you would rely on or provide to ensure compatibility with existing Artemis elements.
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B. Programmatic Topics

1. Provide approaches to sustained high-tempo operations, including manufacturing scalability and reuse strategies.
 2. Propose a phased development and demonstration plan, including uncrewed and crewed demos including ROM costs.
 3. Describe how your system ensures continuous human presence support, with any optional capability for other mission types treated as secondary.
 4. Identify potential non-NASA customers and market synergies.
 5. Explain assumptions regarding contracting approaches, insight/oversight, data rights, mission operations roles, and transitions to service phase operations.
 6. Identify any assumptions regarding the use of Government-owned data, tooling, facilities, expertise, or intellectual property. Describe any such needs and how they would affect development, testing, and operations.
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C. Price and Business Case Topics

1. Respondents should assume that NASA intends to procure recurring transportation services under Firm-Fixed-Price contract structures once technical maturity and demonstrated performance allow. Responses should identify the earliest feasible point of transition to FFP and propose a pathway to achieving it.
2. Recommend a milestone based structure across development phases.
3. Identify cadence, contract quantity, or reuse assumptions needed to close your business case.

4. Describe how your proposed architecture supports long-term affordability and operational viability, including pathways to cost reduction, reuse, increased cadence, and potential non-NASA demand where applicable.
5. Past Performance: Examples of successful implementation in similar environments or use cases.
6. Business Size: Indicate whether your organization qualifies as a small business, small disadvantaged business, or other socioeconomic category under FAR Part 19.
7. Teaming Arrangements: Describe any teaming arrangements, partnerships, or subcontracting plans that would be utilized to deliver the proposed solutions.
8. NASA emphasizes that its Centers (people, hardware, facilities, capabilities, etc.) remain essential national resources. Responses should discuss how proposed concepts leverage, benefit from, or contribute to capabilities at NASA Centers, and what NASA investments (if any) would materially improve schedule, cost, or mission risk posture.
9. Domestic Sourcing: Provide information on the domestic sourcing of materials, components, and labor in compliance with the Buy American Act and other applicable regulations.

4. RESPONSE SUMMARY

Include an Executive Summary of:

- Core human-transport capabilities
- Human spaceflight or hazardous operations experience (robotic experience may be mentioned only as optional background)
- Distinguishing attributes as an established provider or new entrant
- Potential teaming arrangements

5. INSTRUCTIONS FOR SUBMITTAL

Responses must be submitted electronically to:

<https://nasagov.app.box.com/f/39bc1411344d4ea3b91bcf17bd48b4d4>

Responses are due no later than **April 23, 2026, at 11:59 PM EST.**

Only PDF format will be accepted.

Submission Content:

- Organization name, address, and point of contact information
- Per Focus Area:
 - a. Executive summary (maximum 2 pages, 12-point font, 1-inch margins)
 - b. Detailed response (maximum 10 pages, 12-point font, 1-inch margins)
 - c. Appendices may be used for diagrams, tables, or supporting data. (optional, maximum 5 pages)

File Naming Convention: "Organization Name_Lunar TransportationRFI.pdf"

Respondents will receive a confirmation email upon successful submission.

Responses to this RFI will not be shared publicly and will be used solely for internal NASA planning purposes.

Classified export-controlled data should not be included.

6. DISCLAIMER

This is a request for information for planning and information purposes only. This is not a request for proposal or quotation, nor is this a solicitation for a contract or grant award. This RFI does not obligate the Government in any way. The Government will not reimburse the respondents for any costs associated with the information submitted in response to this request.

No corresponding solicitation exists; therefore, do not request a copy of the solicitation. If a solicitation following this is released, it will be posted on www.sam.gov (<https://www.sam.gov/>). It is the interested party's responsibility to monitor this site for the release of any solicitation or synopsis.

Responses to this RFI will be reviewed by both civil servants and support service contractors.

If respondents include proprietary data, NASA may use the submissions for internal planning purposes which may result in future competitive acquisitions

7. POINT OF CONTACT

Questions regarding this RFI may be submitted to:

<https://nasagov.app.box.com/f/39bc1411344d4ea3b91bcf17bd48b4d4>.